

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Supported laboratory operations including equipment qualification (PQ), documentation generation for SOPs and work instructions, and contributing to lab process establishment within a **biologics manufacturing** facility.
- Performed general lab operations, including EM, facility testing, method verifications, and execution of laboratory studies, ensuring the smooth running of daily business tasks.
- Reviewed, trended, and reported on laboratory data to support batch generation, collating and analyzing end-of-quarter and year reports for all raw data.
- Managed routine cleans of laboratory equipment and maintained housekeeping standards, identifying **continuous improvement** opportunities in laboratory processes.
- Responded to standard requests from internal and external partners, escalating issues to management as appropriate and ensuring clear communication.
- Actively participated in functional and cross-functional meetings, supporting internal and external audits and contributing to site safety culture.
- Containerized data analysis scripts using **Docker** and deployed them on **AWS Lambda**, enabling scalable processing of laboratory study results and reducing manual data collation time by **~40%**.

SKILLS

Laboratory Systems – LIMS, E-logs, MODA, Electronic Notebooks

Quality & Compliance – GMP, USP, EP, JP requirements, Equipment Qualification, PQ

Process Analytical Technology – PAT, Method Verification, Data Trending, Data Reporting

Laboratory Operations – SOP Generation, Work Instructions, Consumables Management, Housekeeping

Data Analysis & Scripting – Python, R, SQL, Pandas, NumPy, Scikit-learn

Cloud & Containerization – AWS (EC2, S3, Lambda, CloudWatch), Docker

PROJECTS

Laboratory Data Analysis and Reporting Automation

- Developed Python scripts to automate the extraction, cleaning, and analysis of laboratory study data, processing approximately **1,500 records** per study.
- Created interactive **Power BI** dashboards to visualize key performance indicators and trends from laboratory operations, facilitating faster review by stakeholders.
- Implemented data validation checks using **SQL** queries against a PostgreSQL database to ensure data integrity before reporting, catching over **95%** of common errors.

Biologics Testing Workflow Enhancement

- Designed and documented Standard Operating Procedures (SOPs) for new biologics testing protocols, incorporating **GMP** guidelines and USP/EP requirements.
- Collaborated with laboratory operations teams to streamline workflows, reducing the average time for method verification by **~20%**.
- Supported the introduction of novel testing solutions by preparing samples and executing initial verification runs in a controlled laboratory environment.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)